

Benchmarking Glossary

Key terms and definitions used consistently across the NJ Transit Global Benchmarking Research. Understanding these concepts is essential for interpreting scores, comparisons, and recommendations.

← Rankings Dashboard

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Finance Terms

Technology

Payment Systems

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Ranking Scales

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SG Singapore

CH Zürich

GB London

🇺🇸 NJ Transit

FINANCE & REVENUE TERMS

01 Fare Recovery Ratio Finance

The percentage of a transit agency's total operating costs that are covered by passenger fare revenue. A fare recovery ratio of 60% means fares pay for 60 cents of every dollar spent on operations, with the remaining 40% funded by subsidies, grants, or non-fare revenue.

Why it matters: Several systems in this benchmark are compared on fare recovery, and some readers may confuse this with total cost recovery (which includes subsidies and non-fare revenue). A high fare recovery ratio does not mean a system is independent of government support — capital costs may be publicly funded separately (see: Singapore NRFF, SBB infrastructure subsidies).

Singapore ~55% (operating-cost basis; ~20% govt capital excluded) Tokyo ~58% MTA ~35% NJ Transit ~25–30%

02 Non-Fare Revenue Finance

Revenue not directly collected from passenger fares. Includes advertising, retail concessions, toll revenue, data licensing, property development, payment processing fees, parking, naming rights, and international rail pass sales.

The strategic importance of non-fare revenue is a central theme of this benchmark. Systems with high non-fare revenue are less vulnerable to ridership fluctuations and political pressure on fare levels.

TfL: Congestion Charge, ULEZ JR East: Suica fees, retail SBB: Freight, real estate

03 Subsidy / Government Support Finance

Public funding provided through state, local, regional, or federal sources to support capital or operating costs of a transit agency. Two types are distinct: operating subsidy (covers day-to-day operations) and capital support (funds infrastructure, vehicles, and major projects).

Precision matters here: Singapore LTA is often cited at ~70% fare recovery, but this figure covers only operating costs under a model where the government already owns all rail assets (NRFF) and absorbs all bus revenue risk (BCM). When full government capital support is counted, operating fare recovery is closer to ~55%. This distinction is important: a high operating fare recovery does not mean a system is independent of government support.

NJT: ~68% of ops from subsidy Singapore NRFF: govt owns rail assets SBB: federal infrastructure grants

04 Revenue Diversification Finance

The extent to which a transit agency relies on multiple income streams rather than fares and direct government subsidy alone. A diversified revenue base reduces financial risk and supports investment in service quality independent of political budget cycles.

NJ Transit scores lowest in this benchmark on revenue diversification — it has virtually none of the non-fare revenue streams that sustain peer systems. Revenue diversification is identified as the highest-priority strategic opportunity for NJT.

05 Payment Processing Fees Finance

Revenue earned when a transit-linked payment instrument is used for non-transit purchases and the operator or ecosystem captures a small fee per transaction. Aggregated across millions of daily purchases, these fees generate significant revenue

independent of ridership.

Tokyo's Suica card earns processing fees on every retail purchase at 500,000+ acceptance points. Seoul Metro earns T-money processing fees at 100,000+ retail locations. This model is directly relevant to NJT's potential digital wallet strategy.

Suica (Tokyo) **T-money (Seoul)** **EZ-Link (Singapore)**

06 **Joint Development / Transit-Oriented Development (TOD)** Finance

Real estate development on, above, or adjacent to transit property that generates revenue through leases, land sales, air rights, or increased ridership from new transit-adjacent residents and workers. TOD also increases surrounding property tax revenue for local governments, though transit agencies may capture only a portion of this value directly.

WMATA: 20+ TOD projects **JR East: Lumine retail above stations** **SBB: 800+ station commercial spaces**

PAYMENT SYSTEM TERMS

07 **Open-Loop Contactless Payment** Payment

Payment using a standard bank card or mobile wallet (Visa, Mastercard, Apple Pay, Google Pay) directly at the fare gate or validator, without requiring a transit-issued proprietary card. No registration, no pre-loading, no local card needed.

TfL pioneered open-loop contactless on London buses in December 2012, extending to the full network in September 2014. MTA's OMNY program is the largest open-loop transit rollout in US history. NJ Transit does not currently offer open-loop contactless acceptance.

London OMNY **MTA OMNY** **Singapore SimplyGo**

08 **Closed-Loop Payment** Payment

Payment using a transit-issued proprietary card or digital wallet that is pre-loaded with value and only accepted within the transit ecosystem and designated partner merchants. Requires onboarding by the rider before first use.

Closed-loop systems can evolve into powerful payment platforms if retail acceptance is expanded (e.g., Suica, T-money, EZ-Link). The friction of obtaining a closed-loop card creates onboarding barriers for tourists and infrequent users.

09 EMV **Payment**

Europay, Mastercard, Visa — the global technical standard for chip-enabled bank cards and secure contactless transactions. Commonly used as shorthand for bank-card tap payment in transit. EMV contactless allows riders to tap standard bank cards at fare validators.

Used by TfL, MTA OMNY, Singapore SimplyGo

10 Fare Capping **Payment**

A feature of contactless payment systems that automatically limits the total amount a rider pays within a defined period (daily or weekly), regardless of the number of trips taken. Once the cap is reached, additional trips within the period are free. Prevents riders from overpaying versus a period pass.

MTA's OMNY includes a weekly fare cap equivalent to the 7-day unlimited MetroCard (\$34 as of the benchmarked period). Daily and weekly caps should be stated separately — a common error is listing both as the same figure.

MTA OMNY weekly cap: \$34 **TfL daily and weekly caps on Oyster/contactless**

TECHNOLOGY TERMS**11 GTFS-RT** **Technology**

General Transit Feed Specification – Realtime. A standard format for publishing live transit information, including vehicle positions, arrival predictions, and service alerts. GTFS-RT is the foundation for real-time transit data in third-party apps like Google Maps, Transit App, and Citymapper.

NJ Transit does not currently publish an open GTFS-RT API. Every top-5 system in this benchmark does. This single gap prevents third-party developers from building accurate real-time experiences for NJT riders.

Used by MTA, TfL, Singapore LTA, Seoul Metro **Absent:** NJ Transit

12 **API / Open Data** Technology

An Application Programming Interface (API) is a protocol that allows third-party developers and systems to access transit data. "Open data" means the agency makes such information broadly available — rather than restricting it to internal tools — enabling a developer ecosystem to build rider-facing applications at no cost to the agency.

TfL's 2010 open data decision catalyzed 700+ third-party apps and generated an estimated £130M in annual economic value for London (2017 Deloitte study). Singapore's LTA DataMall is among the first comprehensive multimodal open transit data platforms globally.

13 **Data Licensing** Data

The monetization of anonymized or aggregated mobility data through paid access, commercial licensing, or structured data products. Customers include urban planners, real estate developers, logistics companies, retailers, and researchers seeking insights from large-scale movement data.

Singapore LTA's DataMall is an example of an open data platform that has evolved toward commercial licensing. NJ Transit generates billions of anonymized mobility data points daily and currently earns \$0 from data licensing — one of the most significant missed revenue opportunities identified in this benchmark.

Singapore LTA DataMall Seoul Metro Smart City partnerships NJT: \$0 currently

14 **Real-Time Crowding Data** Technology

Live information on passenger density by vehicle, car, platform, or station used to help riders choose less crowded options before and during their journey. Typically generated by weight sensors in train cars, gate tap counts, or camera-based AI systems.

Seoul Metro's car-level crowding displays — showing real-time passenger density per individual train car before the train arrives — are a documented world first. WMATA and MTA both have passenger counting infrastructure but do not currently publish real-time crowding data to riders.

Seoul Metro: per-car display on platforms TfL: crowding levels in API WMATA: data exists but not published

15 **Silo Integration** UX

The degree to which pre-trip, in-trip, and post-trip services are unified across operators, apps, fares, and data systems. A fully integrated system allows a rider to plan, pay, board, transfer, and review a multi-operator journey without switching apps, re-entering information, or losing data continuity between phases.

This benchmark breaks silo integration into three phases: Pre-Transit (planning, booking, first-mile), During Transit (live updates, crowding, alerts), and Post-Transit (history, loyalty, recommendations). Singapore and Zürich score highest on silo integration; NJ Transit scores lowest.

16 **Pre-Transit / During Transit / Post-Transit** UX

The three phases of the customer journey used to evaluate data connectivity in this benchmark:

Pre-Transit: Planning, booking, ticket purchase, parking, and first/last-mile arrangements before travel begins.

During Transit: Live status, real-time crowding, delays, alerts, connections, and onboard or in-station information during the trip.

Post-Transit: Trip history, receipts, loyalty, recommendations, feedback, and service recovery after the trip ends.

No US system in this benchmark achieves "Fully Connected" on all three phases. Singapore and Zürich do.

17 **First/Last Mile Integration** UX

Integration of transit with rideshare, bikes, scooters, parking, and walking connections needed to complete the trip from the rider's origin to the transit station, and from the destination station to the rider's final destination. The "first mile" and "last mile" are often the most friction-laden parts of a transit journey.

Singapore: bike-share booking in MyTransport app **SBB:** car-share in SBB Mobile **LA Metro:** first/last mile gap is primary UX challenge

18 **Unified Digital Front Door** UX

A single app or digital interface that lets riders plan, pay, travel, and review their complete journey without switching between systems or tools. The "front door" metaphor captures the idea that riders should have one entry point to the entire transit ecosystem, regardless of how many operators, modes, or agencies are involved in their trip.

19 **Personalization Layer** UX

Use of rider journey data to tailor suggestions, alerts, loyalty offers, disruption alternatives, and post-journey recommendations to the individual rider. A personalization layer turns anonymized mobility data into a rider-facing benefit — proactive journey suggestions, relevant alerts, and rewards aligned with actual travel patterns.

The absence of a personalization layer is a common gap across US systems in this benchmark. OMNY, SmarTrip, and TAP all generate rich journey data that is not currently surfaced to riders in personalized form.

20 **Customer Centricity** UX

The degree to which a transit system is designed around the rider's complete journey experience, rather than the operational or administrative boundaries of individual agencies and operators. A customer-centric system prioritizes the rider's perspective across all touchpoints: planning, payment, travel, and follow-up.

Singapore and Zürich score highest on customer centricity — their systems are designed so that operational complexity is hidden behind a simple rider experience. NJ Transit's "three-app problem" is the clearest benchmark expression of low customer centricity: the system's internal fragmentation is directly visible and painful to the rider.

RANKING SCALE REFERENCE — WHAT 100 AND 0 MEAN FOR EACH DIMENSION

24 **Technology Stack Score (0–100)** Scale

Measures the depth and maturity of a transit system's technical infrastructure: real-time data publishing (GTFS-RT), open APIs, mobile ticketing, contactless payment acceptance, AI or predictive tools, and interoperability with third-party platforms.

Score	What it means	Benchmark example
100	Full real-time open API suite (GTFS-RT, vehicle positions, alerts), open-loop EMV contactless across all modes, AI-driven disruption prediction, seamless third-party app integration, and a fully interoperable tech platform with no proprietary barriers.	No system in this benchmark scores 100. Singapore LTA scores 97 — the closest to this ceiling.
0	No real-time data published externally, no contactless payment, no mobile ticketing, no open APIs, no third-party app support. Riders have access only to static printed schedules and cash fare payment at windows.	No benchmarked system reaches 0. NJ Transit (45) has the lowest score, primarily due to the absence of open GTFS-RT and no contactless fare payment.
Singapore: 97 Zürich: 94 London: 92 Seoul: 90 MTA: 68 NJ Transit: 45		

25 **User Experience (UX) Score (0–100)** Scale

Measures how easy and unified the rider experience is — from planning a trip, to paying, boarding, receiving real-time updates, and recovering from disruptions. Rewards systems where operational complexity is hidden behind a simple, consistent interface.

Score	What it means	Benchmark example
100	A single app handles planning, payment, live updates, and post-journey review across all modes and operators with zero app-switching. Accessibility, language support, wayfinding, and disruption recovery are fully integrated. Every touchpoint is intuitive and consistent.	Zürich SBB scores 96 — riders plan, book, pay, and receive automated delay compensation in a single app across all Swiss national rail and urban transit.
0	No digital rider tools exist. No app, no website trip planner, no real-time information at stations, cash-only payment, and no disruption communication beyond in-station announcements. Riders must navigate entirely without digital support.	No benchmarked system is near 0. NJ Transit (42) is the lowest, characterized by 3–4 separate apps, no unified payment, and absent post-journey features.
Zürich: 96 Singapore: 95 Tokyo: 90 Seoul: 88 MTA: 65 NJ Transit: 42		

26 **Silo Integration Score (0–100)** Scale

Measures how connected the three journey phases are — Pre-Transit (planning & payment), During Transit (live updates & alerts), and Post-Transit (history, loyalty & personalization) — both within one agency and across multiple operators sharing the same network. Also expressed qualitatively: **Best-in-class / High / Medium / Low**.

Score	What it means	Benchmark example
100	All three journey phases are fully connected in a single platform. Pre-trip planning, real-time in-journey data, and post-journey loyalty are unified across every mode and operator. Data flows seamlessly between phases; no information is lost or siloed when the rider transitions from planning to travel to review.	Singapore scores 98 — MyTransport.SG unifies planning, payment, live updates, and journey history across all bus, MRT, and LRT operators. Qualitative label: Best-in-class .
0	Completely siloed: no data passes between pre-trip planning, the journey itself, and post-journey review. Riders must re-enter information at every phase, alerts are absent during travel, and no record of trips exists afterward. Each operator is an isolated data island with no cross-mode connectivity.	No benchmarked system reaches 0. NJ Transit scores 30 — the lowest — with no open real-time API, absent post-journey features, and operations split between NJT and Amtrak infrastructure. Qualitative label: Low .

Singapore: 98 — Best-in-class Zürich: 95 — Best-in-class London: 90 — High MTA: 62 — Medium NJ Transit: 30 — Low

27 Revenue Model Score (0–100) Scale

Measures how diversified and self-sustaining a transit system's revenue model is. Rewards systems that generate significant income beyond passenger fares and direct government subsidy — including payment processing fees, retail concessions, property development, data licensing, freight, advertising, and congestion or road pricing revenue.

Score	What it means	Benchmark example
100	A fully diversified revenue model where fares are one income stream among many. The agency generates substantial revenue from retail concessions, payment platform fees, real estate development, data licensing, freight operations, and other commercial activities. The system can sustain and invest in operations independent of political budget cycles, with non-fare revenue covering 50%+ of operating costs.	No system scores 100. Tokyo JR East (92) comes closest — non-rail revenue (Suica processing fees, Lumine retail, hotels, real estate) exceeds 40% of total revenue.
0	100% dependent on passenger fares and direct government subsidy with zero non-fare commercial revenue. No advertising, no retail, no data licensing, no property income, no payment fees — every dollar	No benchmarked system reaches 0. NJ Transit (50) is the lowest, earning virtually no non-fare commercial revenue. Its budget relies ~70% on

Score	What it means	Benchmark example
	comes from the farebox or a government transfer. Completely vulnerable to ridership drops and political budget decisions.	subsidies and ~25–30% on fares, with no payment platform, data licensing, or meaningful property revenue.
Tokyo: 92	Singapore: 90	London: 88
Zürich: 85	MTA: 75	NJ Transit: 50

BENCHMARK METHODOLOGY

21 **Benchmark Score** Methodology

A comparative score assigned within this research framework across four dimensions: Technology Stack, User Experience, Silo Integration, and Revenue Model. Scores are on a 0–100 scale and reflect the research team's assessment based on publicly available information, agency reports, and direct product evaluation.

***Important:** These scores are internal to the NJ Transit Benchmarking Research 2024 study. They are not independently validated external industry ratings, and should not be cited as such. The methodology weights each dimension equally; no external tie-breaking methodology is applied.*

22 **New Rail Financing Framework (NRFF)** Governance

Singapore's government framework, introduced in 2010, under which the Land Transport Authority (LTA) owns all rail operating assets — trains, tracks, and signaling equipment — and operators pay a license fee to run services. This means the government bears all capital risk, and operators focus purely on service delivery.

The NRFF fundamentally changes how to interpret Singapore's "high fare recovery": fares cover operating costs in a model where the government has already absorbed the capital investment. Singapore's financial model is not as subsidy-independent as a standalone fare recovery figure suggests.

23 **Bus Contracting Model (BCM)** Governance

Singapore's bus financing model, introduced in 2016, under which the government (through LTA) bears all revenue risk — setting fares and absorbing fare volatility — while bus operators are paid a fee to operate services to quality standards. This

model insulates operators from ridership risk and gives LTA full control over network planning.

Together with the NRFF, the BCM means Singapore's government is deeply involved in financing public transit, even though the systems present high fare recovery ratios. This context is essential for fair cross-system comparison.

SOURCES FOR GLOSSARY DEFINITIONS

Fare Recovery / Revenue definitions: WMATA FY2024 Adopted Budget; MTA FY2024 Budget; TfL Annual Report 2023/24; JR East Annual Report; Seoul Metro Annual Statistical Report; Singapore LTA Annual Report; SBB Annual Report; LA Metro FY2024 Adopted Budget; NJ Transit Annual Report.

Technology definitions: Google GTFS-RT specification (developers.google.com/transit/gtfs-realtime); TfL Open Data Portal (api.tfl.gov.uk); LTA DataMall (datamall.lta.gov.sg); MTA Developer Resources (api.mta.info).

Payment systems: EMV specification ([emvco.com](https://www.emvco.com)); TfL contactless history (tfl.gov.uk/corporate/publications-and-reports); MTA OMNY program (omny.info).

Singapore NRFF/BCM: Singapore Land Transport Authority — New Rail Financing Framework documentation; Bus Contracting Model implementation reports.

Open data value: "Assessing the value of TfL's open data and digital partnerships," Deloitte for TfL, 2017.

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